Thermodynamics and Statistical Physics

5th Exercise Class

Entropy; Clausius inequality, Fundamental thermodynamic relation

Exercises for the Class

1. 5 kg of water at 25 °C is added to 10 kg of water at 85 °C. After the mixture has reached equilibrium, how much has entropy changed? (Assume no energy is exchanged between the water and its surroundings. The average specific heat of water between 0 °C and 100 °C is 4.18 J/(g°C).)

2. Calculate the entropy change for each of the following:

   (a) 10 g of steam at 100 °C and pressure of 1 atm condensing into water at the same temperature and pressure. (The latent heat of vaporization of water is 2260 J/g.);

   (b) 10 g of water at 100 °C and a pressure of 1 atm cooling to 0 °C at the same pressure. (The average specific heat of water between 0 °C and 100 °C is 4.18 J/(g°C).);

   (c) 10 g of water at 0 °C and a pressure of 1 atm freezing into ice at the same pressure and temperature. (The latent heat of fusion of ice is 333 J/g.)

3. Consider the heating of a beaker of water with heat capacity $C_p = 5$ kJ/K, from $T_i = 293$ K to $T_f = 373$ K. For the subquestions consider Fig. 1.

   (a) Heating is carried out irreversibly, via heat transfer from a reservoir at $T_\mathrm{r} = 100$ °C. The reservoir is large enough that its temperature does not change in the process, see Fig. 1a. Calculate the change of the entropy of the whole system (water and reservoir).

   (b) Heating is carried out irreversibly, via two reservoirs: one at $T_{\mathrm{r}1} = 50$ °C and one at $T_{\mathrm{r}2} = 100$ °C. The reservoirs are large enough that their temperature do not change in the processes, see Fig. 1b. Calculate the change of the entropy of the whole system (water and reservoirs).

   (c) Heating is carried out reversibly, via a Carnot engine between a reservoir at temperature $T_\mathrm{r} = 100$ °C and the water, see Fig. 1c. Calculate the change of the entropy of the whole system (water, reservoir and Carnot engine).

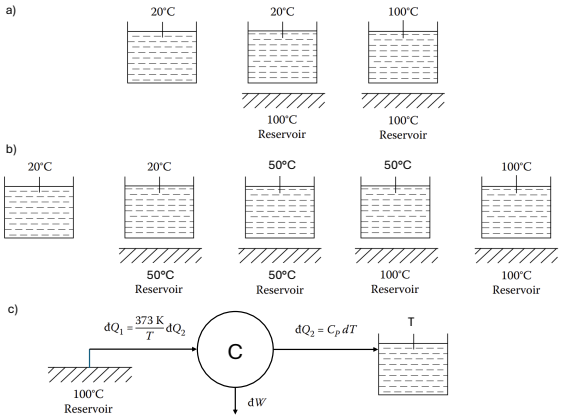


Fig. 1

4. A gas has a molar specific heat given by $c_V = = A + BT$ where A and B are constants. Show that the change in entropy per mole going from the state (V_1, T_1) to the state (V_2, T_2) is

$$\Delta S = A \ln\left(\frac{T_2}{T_1}\right) + B(T_2 - T_1) + R \ln\left(\frac{V_2}{V_1}\right)$$

   Consider that the equation of state for ideal gas at constant temperature is valid for this gas as a good approximation.

5. A thermally insulated container is divided into two equal volume sections by a light, thermally insulated piston, which is initially fixed in place. On the left side, there is a quantity of n moles of a diatomic ideal gas at temperature T_0, and on the right side, there is a vacuum. At a certain moment, the piston's fixation is released. ($\gamma = 7/5$)

   (a) What will the temperature in the container be after a long time and how much has the entropy changed?

   (b) The piston is slowly pushed back to its original position, and then fixed again. What will the gas temperature be in this new situation?

   (c) How much work was done during the process of pushing the piston back?

   (d) How much has the entropy of the gas changed?

Homework to Be Submitted

1. Consider a system consisting of a bar, with heat capacity C_p, inside an adiabatic enclosure, with one end initially hotter than the other, as in Fig. 2. Assume that the temperature depends linearly on the position. After long enough time, the temperature equalizes and the gradient vanishes. Calculate the entropy change during the process. Check that the entropy change is positive.

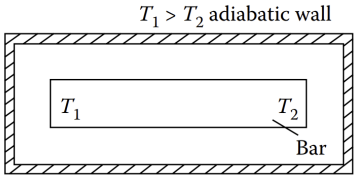


Fig. 2

Homework for Practice

1. Water with mass of $2m$ and temperature of T_0, and water with mass of $3m$ and temperature of $2T_0$ are poured into a thermally insulated container. The specific heat of water is c.

   (a) Determine the equilibrium temperature T_1 after thermal equilibrium is reached.

   (b) Determine the change in entropy of the system. Did the entropy increase or decrease during the process?

   (c) Next, we mix $5m$ of water at temperature T_1 with water in a large heat capacity heat reservoir at temperature T_0. Determine the entropy change of the entire system (water and reservoir combined). Did the entropy increase or decrease during the process?

   (d) What should the temperature of the heat reservoir be if we want the entropy change of the entire system (water and reservoir combined) to have the opposite sign compared to case (c)?

2. During a quasi-static process involving 2 moles of an ideal gas with an adiabatic exponent of 1.3, the volume doubles and the pressure decreases to one-third of its original value. By how much does the entropy of the gas change?

3. We transfer one mole of a van der Waals gas from state (V_1, T_1) to state (V_2, T_2). Determine the change in entropy of the gas, given that its specific heat at constant volume is C_V!

4. The internal energy and pressure of a photon gas are given by the following equations:

$$U = aVT^4,$$
$$p = \frac{1}{3}aT^4,$$

   where a is a constant. Determine the entropy of the photon gas.